



Teaching in Higher Education

Critical Perspectives

ISSN: 1356-2517 (Print) 1470-1294 (Online) Journal homepage: <https://www.tandfonline.com/loi/cthe20>

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Michael Kwet & Paul Prinsloo

To cite this article: Michael Kwet & Paul Prinsloo (2020): The 'smart' classroom: a new frontier in the age of the smart university, Teaching in Higher Education, DOI: [10.1080/13562517.2020.1734922](https://doi.org/10.1080/13562517.2020.1734922)

To link to this article: <https://doi.org/10.1080/13562517.2020.1734922>



Published online: 14 Apr 2020.



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The 'smart' classroom: a new frontier in the age of the smart university

Michael Kwet^a and Paul Prinsloo ^b

^aYale Law School, Yale University, New Haven, CT, USA; ^bDepartment of Business Management, University of South Africa, Pretoria, South Africa

ABSTRACT

This article examines developments in the 'smart classroom' as a new frontier for the university. It provides a conceptual map of the scope and limitations of smart classrooms, contextualized to smart university initiatives. First, it introduces the notion of 'smart' technology in cities, campuses, and classrooms. Next, it examines how the smart classroom is conceptualized, its present applications, and how it may be utilized with or without integration into the smart campus. Finally, it explores the ethical implications of the smart classroom. It concludes by recommending new regulations for smart classrooms and provides examples of classroom technologies constructive to the pedagogical process.

ARTICLE HISTORY

Received 31 July 2019

Accepted 23 February 2020

KEYWORDS

Smart classroom; smart campus; smart university; big data surveillance; education technology; smart city; digital colonialism

Introduction

Technology, throughout the ages, has impacted on and shaped social life. The digital revolution (Kitchin 2014a) is no exception, and has unleashed new technological capabilities which are altering the fabric of society at a speed and scope never seen before. Within a few decades, computers have evolved from mainframes to household items, to personal mobile devices, and now to small objects for the Internet of Things (IoT). Physical items such as toasters, toothbrushes and even diapers are becoming 'smart' devices driven by computer technology (Greengard 2015). So, too, are physical environments: the 'smart city,' we are told, will utilize a combination of 'sensors, actuators, meters, transponders, cameras, and computational devices' as well as 'crowd-sourcing and locative and social media' in order to understand and administer city life (Kitchin 2014b; Goldsmith and Crawford 2014).

Higher education institutions are moving along a similar trajectory. While technology in education has always influenced not only *what* is taught but also *how* it is taught, technologies like artificial intelligence are increasingly shaping knowledge and research – the very foundation of education. Education technology is therefore shifting from being predominantly pedagogical tools and a means for access to knowledge to realizing Big Data surveillance for 'personalized' education based on the collection of data from individual student devices (Watters 2014; Daniel 2015). As learning analytics

has become more commonplace (Gašević, Dawson, and Siemens 2015; Siemens 2013), we increasingly see efforts to intensify and expand the collection of student data in what Beer (2019) describes as a ‘data frontier.’ Today, higher education institutions are extending the data frontier through ‘smart campus’ and ‘smart classroom’ initiatives that utilize many of the same technologies and concepts found in smart and safe city projects. As such, physical space, and the broader project of student data collection, is becoming part of an ‘elaborate lattice of information networking’ (Solove 2004, 3) where data are shared, sold, merged, and new data bodies and data profiles come into existence (Jones 2019).

Smart classrooms and smart campuses are distinct phenomena. In this article, both terms refer to the use of IoT systems, cameras, sensors, and other computational devices that conduct physical surveillance of education participants in their environment to manage educational processes. The *smart classroom* includes online and offline classroom spaces, whereas the *smart campus* includes broad campus activity. As we will see below, the smart classroom and the smart campus can intersect, or they can be deployed in isolation from each other.

Literature on smart classrooms tends to focus on technical systems and the technical feasibility of deployment (e.g. Timms 2016; Kim, Soyata, and Behnagh 2018; Popescu et al. 2018; Satu et al. 2018; Abdellatif 2019). Publications critical of smart classroom technologies such as affect recognition (e.g. Williamson 2017a, 2017b; Watters 2017) are scarce, the topic is underdeveloped, and there is little empirical research gathering data from project architects and implementers.

This article contributes an early snapshot of smart classroom initiatives. It offers a conceptual map of the scope and limitations of the information lattice between the smart city, the smart campus, and the smart classroom. It also provides empirical findings on smart classroom projects from the scholarly literature and two original field interviews – one from a major university, the University of Arizona, the other from a widely publicized emotion recognition company, Nestor AI. The article also analyzes the ethical and practical implications for the institution of higher education. It concludes with recommendations for how to proceed with smart classroom technology in education.

The rise of ‘smart’ environments in the data imaginary

The notion of a ‘data imaginary’ is constituted by the accepted, dominant, and unquestioned narratives pertaining to how Big Data might improve society in any particular context. Much like the social imaginary (Castoriadis 1987), it is a ‘generative matrix’ that *legitimizes* ‘practices and pursuits and embed them in a normative scheme’ (Gaonkar 2002, 10). Williamson (2015) links the data imaginary back to the notion of a ‘sociotechnical imaginary’ that presents a ‘publicly performed vision of a desirable future’ that is ‘made attainable through the design of technological projects’ (2). Particular social groups produce these visions ‘to express a view of particular futures in which those kinds of technologies are imagined to be integral, embedded parts’ (Williamson 2015).

Such data imaginaries are generated by those who have the ‘power to speak with those data,’ or, ‘who [have] the power to speak with our data’ (Beer 2019, 1). Williamson (2015) links the data imaginary to ‘dreamscapes of the future’ constructed by Big Tech proponents, which gives a techno-determinist ‘sense of inevitability’ to Big Data surveillance

(Dourish and Bell 2011, 2). Any questioning of the narrative or technotale/myth is made out to be ‘against progress, a Luddite or worse’ (Dourish and Bell 2011).

In order to be ‘smart,’ technologies and policy processes often require massive amounts of information to personalize experiences and power artificial intelligence systems. In the emerging data imaginary, our economies, policies, buildings, cities, houses, cars, and myriad other ‘things’ will be ‘datafied’ so they can be analyzed with computers. Such computational transformations of pre-digital society bake mass commercial and government surveillance into the things and processes of everyday life (Schneier 2018; DeNardis 2020).

The data imaginary, shaped by Western intellectual and corporate narratives, serves to ‘manufacture consent’ (Herman and Chomsky 1988/2002) around the extension of Big Data to the global society through the process of digital colonialism (Kwet 2019a, 2019b). What can be accomplished with data is usually a euphemism for what can be accomplished with surveillance, because Big Data is typically needed to assess human behavior with predictive accuracy. The data imaginary is thus a manifestation of *tech hegemony* to make our computer-driven societies accept the idea of being ‘smart’ (Kwet 2019a, 2019b).¹

Physical environments like cities are said to become smart when they begin to merge IoT and Big Data. Smart universities are building smart environments intimately tied to developments in smart cities. With smart cities, governments and corporations ingest massive troves of data from widespread sensor and CCTV networks, utility services, social media, public records, and other sources, to understand and manage life in the city (Kitchin 2014b). By making the city knowable in dynamic and granular fashion, the efficiency, sustainability, and quality of city life can allegedly be predicted and enhanced, based on a neoliberal public-private partnership approach to planning and implementation (Grossi and Pianezzi 2017).

Smart city data assemblages resemble information centralization practices developed by city police departments in the United States. During the early 1990s, several New York Police Department (NYPD) officers developed CompStat (‘comparative statistics’) to inform and predict policing practices for patrol units. CompStat soon evolved into Real-Time Crime Centers (RTCCs), which collect numerous streams of data for Big Data policing (Ferguson 2017). The integration of information into sophisticated surveillance software systems has given rise to tech-driven ‘safe cities’ initiatives said to provide law enforcement agencies unprecedented understanding of crime and social patterns for data-driven policing (Kwet 2020).

Smart cities extend the logic of RTCCs to broad city life. While some innovations promise to improve waste management, lighting, energy efficiency, and traffic flow (Goldsmith and Crawford 2014), critics charge the deployment of sensors, IoT networks, and CCTV cameras centralize power for technocratic management and corporate profit (Hollands 2008; Kitchin 2014b; Privacy International 2017). Tech-driven smart cities thus deepen surveillance, privatize urban governance, and undermine democracy (Mosco 2019).

Universities are valuable sites to work out new smart city solutions, often in collaboration with private corporations. College campuses are small and relatively self-contained environments (Fortes et al. 2019), while university personnel offer human resources for experimentation and collaboration. Much like smart cities and RTCCs, smart campuses seek to combine an array of data from disparate sources for analytics to modify the

traditional university experience. Applications could include environmental sustainability and energy efficiency, campus life management such as wayfinding or event organization, sensor deployment for coordination of facilities, IoT and CCTV-based video analytics for location analytics and security, smart classroom analytics such as affective learning, smart identification cards for services and building accessibility, and integration of data sources for university performance outcomes like student retention and learning.

Perhaps the most ambitious smart campus pilot project is in motion at Curtin University in Australia. An initiative launched in 2017, Curtin's smart campus is utilizing sensors, CCTVs, and Big Data surveillance for the total university experience. 'Gathering information on building trends, study patterns, course attendance, and social interaction can ultimately be used to improve student experience and enhance learning,' Professor Lina Pellicione, Head of Curtin's School of Education, remarked in a promotional video. 'The data allows us to generate contextual information about the life cycle of the student, the day-to-day reality of the staff member, the activity pattern of a lecture theater, and the dynamics and environmental health of a library' (Hitachi 2017).

Curtin's initiative exemplifies the social imaginary of what it means to be a smart university in the digital era. Building 'a campus of the future' means having 'the courage to invest in doing things differently at a scale.' Competition frames their rationale: Curtin is 'rising in rankings across the world,' and must be 'innovative ... dynamic and ... competitive' to leap ahead of the curve in a zero-sum game against other universities (Cisco 2017).

Smart campuses are thus rooted in the neoliberal imperative to quantify behavioral patterns and compete for better numbers with respect to throughput, grades, and other educational performance metrics. As one of the most thorough-going smart university projects in the world, Curtin's aggressive approach to technology integration – based on CCTV cameras, Internet monitoring, and IoT for centralized analytics – represents one end of a spectrum. Yet there is no single smart campus model, and universities differ in their approach to technological integration. Some campuses in the United Kingdom are dipping their toes in the water by piloting new technologies in a more limited and piecemeal fashion. For example, Bolton College is deploying an AI-assisted chatbot for use on Amazon Echo devices that pulls details about students into a university data repository, while Sheffield Hallam University is deploying Bluetooth beacons for wayfinding with a smartphone app for use in libraries and information kiosks.

Smart classrooms: a new frontier for higher education

EdTech advocates believe that with enough data, they can assess people in ways useful to education. Their ideology expresses a frontier logic that seeks to extend Big Data surveillance to the field of education (Kwet 2019b). Learning analytics aims to fulfill the decades-old dream to 'personalize' learning for individual students, often using one-to-one intelligent tutors that cater to students on an individual basis (Pardo et al. 2019). Other applications for learning analytics include early alerts for students deemed at risk of failure or dropping out, course recommendation, and curriculum design (Sclater 2017, 33–76). Data can also be utilized by administrators for applications like longitudinal studies that track the performance of students, teachers, departments, and universities over time (Ferreira 2015).

While the notion of the ‘classroom’ originated in traditional education settings, it is also used in the context of distributed learning environments, whether online, blended, and/or distance learning. These forms of educational delivery often differ not only with regard to student demographics, costs, retention, and drop-out, but also pedagogical strategies, admission requirements, and flexibility (e.g. Shachar and Neumann 2010; Subotzky and Prinsloo 2011). While education (no matter the delivery format) has always used a variety of data (e.g. demographical and behavioral data) for planning, operational, and pedagogical purposes, smart technologies offer new opportunities to expand the ‘data gaze.’ As such, smart classrooms offer not only new forms of data, but also a greater variety, granularity, and velocity of data that can be used, inter alia, to inform teaching and learning. Cameras and sensors are typically used to transform or alter the learning environment for a range of purposes, including informing pedagogical strategies, by way of surveillance, monitoring, and sharing outside of the original context of collection. Smart classrooms can stand on their own, or they can be integrated with broader smart campus and university initiatives.

It is crucial to note that data collected in these forms of delivery may differ depending on technologies, context, institutional character, and pedagogical strategies; that the data are proxies for learning; and that the data are not only incomplete and contingent, but often ‘broken’ (Broughan and Prinsloo 2019; Pink et al. 2018). Student data also flow from, constitute, and perpetuate existing and new forms of asymmetrical power relations between students, educators, administrative staff, and increasingly, external providers and data brokers. Moreover, there are increasing concerns that the smart technologies may perpetuate and exacerbate historical and current inequalities (Lim and Tinio 2018). In the following sections we will explore smart classroom developments in the physical and online experience, as well as integration into the smart university, before turning to ethical considerations.

Smart classrooms for physical classes

Smart classrooms use cameras, sensors, smart devices, and IoT systems to collect new sources of data about student participation (e.g. raising hands and discussion), facial expressions, bodily postures, and engagement (through the use of products like Clicker). Such information, proponents hope, will generate new insights about individual and group behavior inside the classroom.

Affect analysis is a popular form of analytics that assesses emotions and moods expressed in smart classrooms (D’Mello 2017). *Emotional states* are thought to perform functions that signal problems such as boredom, confusion, anxiety, and frustration. Emotions play a critical role in learning because they impact motivation, attention, and concentration (Arroyo et al. 2009). As such, learning analytics and education data mining are thought to be incomplete without emotional data (D’Mello 2017, 116).

Affective states, by comparison, encompass both emotions *and* moods (Gupta, Ashwin, and Guddeti 2019, 8–9). *Affective computing* is defined as ‘computing that relates to, arises from, and influences emotion’ (Picard 1995, 1). Using video analytics, sensors, and other sources of data, the field of affective learning seeks to measure a student’s ‘affective and cognitive states ... in order to increase the motivation, attention and engagement of that student’ (Popescu et al. 2018, 1).

According to Popescu et al., ‘teaching staff must not only teach’ but also ‘monitor’ and ‘know the status of their students during a lab’ so they can adapt their teaching to ‘the emotional and cognitive state of the students’ (Popescu et al. 2018). The authors recently developed a low-cost system comprised of sensors to measure chair pressure (for sitting posture), heart rate, and galvanic skin response, as well as web cameras (to detect facial expressions). Cognitive states profiled include attention/curiosity, engrossed/engaged, confused, and distracted.

Popescu et al.’s prototype increases affect recognition accuracy by utilizing multimodal emotion detection, which alleviates the constraints of individual measurement channels (Popescu et al. 2018, 232) by encompassing more facets of emotions (Kapoor and Picard 2005; Arroyo et al. 2009; Kim, Soyata, and Behnagh 2018). However, in their system, students cannot move around very much and must remain fixed to their computers, which places practical limitations on their solution.

Other smart classroom models make use of more free-flowing technologies – RFID tags, Bluetooth Low-Energy (BLE) beacons, routers, sensors, cameras, microphones, mobile devices, and wearables – for data collection into local or cloud-based databases. Using these technologies, researchers claim visualization and analysis tools can perform educational tasks that ‘[generate] alerts and guide the teacher during the session’ (Abdelatif 2019, 282).

Satu et al. (2018) propose a simple combination of cameras, microphones, and sound sensors to monitor the learning process and identify problems of students (3). Timms (2016) suggests adopting wearable badges that evaluate emotion from the affective tone of the voice, as well as biometric wristbands and motion sensors to gather information about student engagement or alert teachers to students off task in group work. Timms also suggests the use of high-resolution facial recognition cameras to detect student emotions and data mine productive classrooms to build predictive models for classroom environments.

Intel partner Corerain’s ‘artificial intelligence smart classroom’ solution uses video analytics to detect when students express ‘positive actions’ (such as ‘participation,’ ‘hand raising,’ and ‘standing up’) and ‘negative actions’ (such as ‘turning around’ and ‘resting their head on the table’). These actions are identified and counted for duration, where they can be logged and visualized in a dashboard to determine an engagement model. As Intel explains (Intel 2019), ‘The classroom video is turned into Big Data analytics’ that ‘can evaluate the teaching quality of teachers, and help them to adjust their teaching methods to increase student engagement’.

In a research interview, Arizona State University’s Chief Technology Innovation Officer, Charles Kazilek, said he is leading a smart classroom experiment that utilizes emotion recognition, location, and facial recognition. As a ‘sandbox’ classroom for experimentation, it will use 360 degree cameras and other equipment – including Sony’s face recognition technology – for insights that could guide pedagogical practices in an active learning classroom. He said that video analytics will help discern if students look puzzled, understand material, or seem to be enjoying their work. It will also keep track of student movements to evaluate what happens when they cluster: ‘Is it better to have students in pairs, or three, or four? All these types of things’ (Kazilek 2019).

The technology will allow faculty to see what the students are doing in the learning space ‘and then overlay that with content that is coming in [with respect to] student

achievement’ (Kazilek 2019). Data collected will be uploaded into the learner management system (LMS) to synthesize with other learner analytics. Comparisons will be made to a control group – another classroom teaching the same content that does not utilize smart classroom technology – which Kazilek hopes will provide early insights about whether or not the technology is improving the educational process.

Smart classrooms for E-learning classes

E-learning, or online learning, refers to a range of practices. These include fully online offerings with no physical interaction, blended options (whether physical or fully online), and distance education courses with some online interaction in support of printed materials (Johnson et al. 2019). Some of the smart technologies discussed in the previous section (for physical classroom settings) are equally applicable in e-learning classes.

E-learning courses are commonly integrated with LMSs to allow students and teachers to collaborate, communicate, and coordinate class work and activities online. LMSs are also often used to surveil educational activities in fine detail (Johnson 2019). Studies suggest that students perform worse in online courses at universities, with negative effects concentrated in the lowest performing students (Bettinger and Loeb 2017).

Intelligent virtual tutors seek to enhance online education by assessing student strengths and weaknesses and diagnosing why students make specific errors. To improve online learning outcomes, some education technologists want to utilize affect recognition.

One such learning system, Nestor AI, made splashes in the popular press for its unrestained approach to student analytics. One promotional video shows a learner playing with his cell phone as an online lecture begins. A giant hand reaches out of his laptop and smacks the learner in the face: ‘You are not attentive,’ the robot exclaims. The hand represents Nestor AI’s analytics engine, which uses face detection in real time to determine his state of attention. Warnings are sent when the student becomes distracted, and quizzes are formulated based on the content covered during moments of inattentiveness. Nestor AI also claims it can use data from social networks and register missing students from class (Nestor 2017).

In a research interview, Marcel Saucet, founder of LCA Learning, the company that created Nestor, said that the software recognizes over 400 facial movements and assesses the content of student keyboard inputs to determine if students are typing about the class material (rather than something unrelated to their online coursework). Saucet would like Nestor AI to connect learning accounts to the students’ Gmail, YouTube, and Google Docs in order to ingest information such as what time students watch videos and the length of the videos. By ‘collecting enough information,’ such as data about your facial expressions across many classes, the software will ‘know the students really well’ so it can perform predictive analytics – like when students are likely to stop paying attention – on a personalized basis. Nestor AI solutions can be either hand-tailored or integrated into e-learning platforms like Moodle or Blackboard (Saucet 2019).

While Nestor AI is for online use with webcams, Saucet said it would be good to have cameras in a lecture hall ‘following the movements’ of students like ‘Big Brother.’ Nestor, he explained, would generate reports for professors. If just two students weren’t paying

attention, then those students would likely be the ones creating the problem. But if 200 students were inattentive, ‘then the problem is with [the teacher’s] pedagogy, or his slide, or whatever,’ Saucet remarked. He added that current deployments advertised elsewhere in-classroom would differ in terms of the level of detail they want to achieve. Given that Nestor AI is built to distinguish the minute details of about 400 facial variables, the software would have to be redesigned for a camera observing a large classroom, which does not capture such granular detail (Saucet 2019). Saucet said his company is not currently selling student data, but added that if ‘in five years, I have billions of dollars of data, maybe I will change my mind’ (Saucet 2019).

Other smart online classroom experiments use webcams for similar tasks. Pilots vary according to factors like the type of emotions measured, duration and temporal dynamics of affective states, factors that induce emotions, and the methods used for detection, generation, and regulation of emotions (Azevedo et al. 2016; for models and projects, see, *inter alia*, Bahreini, Nadolski, and Westera 2016; Kim, Soyata, and Behnagh 2018). Models differ considerably, and there is no consensus on which methods of measurement are the most effective (Harley 2016), let alone which systems work the best for pedagogical purposes. This suggests smart e-learning classrooms are highly experimental, and there is no clear winner for the intelligent tutoring environment.

Smart university integration and mission creep

In order to improve the quality of their analysis, smart classroom researchers seek to widen the scope of data collected. Such expansionary impulses cover behavior inside and outside the classroom. Why stop at facial expressions when you can also have data on body language? Why limit yourself to the classroom when you can know the broader life patterns of student behavior?

A recent study commissioned by the Association for Psychological Science illustrates the limitations of in-class emotional analysis. After reviewing 1,000 different studies on emotion and facial recognition, the authors concluded it is difficult to reliably infer emotion from a simple set of facial movements because there is no strong correlation between facial expressions and emotions (Barrett et al. 2019). For instance, people scowl less than 70 percent of the time they are angry, and a scowl may express emotions other than anger. Multimodal forms of sensing are thus needed to accurately detect emotions. As detailed above, this could correlate with device behavior (such what a person is typing) or other sensory information (such as galvanic skin response or sitting posture). The reliable assessment of emotions with computers therefore requires increasingly intrusive forms of surveillance.

Indeed, according to D’Mello (2017), thanks to advances in computer technology and data analytics, researchers are devising new ways to evaluate teachers with machines. For instance, Wang, Miller, and Cortina’s (2013) ‘pioneering study’ used audio recordings to predict the level of discussions in 1st and 3rd grade classes. Their work was recently expanded to include instructional activities, such as small group work or supervised seat-work, using teacher audio alone (Donnelly et al. 2016a), or a combination of teacher and classroom audio (Donnelly et al. 2016b). D’Mello, a distinguished researcher in the field of affective computing, states, ‘The next step in this line of work is to use information on what teachers are doing to contextualize how students are feeling, which in turn influences

what they think, do, and learn' (D'Mello 2017, 121–122). Teachers can be subjected to the data gaze directly, through physical teacher surveillance, or indirectly, if students' affective states and movements are linked to learning outcomes and viewed as a reflection of teacher performance.

For D'Mello, 'emerging themes ... at the intersection of emotions, learning, [learner analytics], and [educational data mining]' include teacher analytics, affect-based predictors of attrition and dropout for early intervention systems, sentiment analysis in discussion forums, and classroom learning analytics. 'Future themes' include emotional tracking across the extended time of a class, emotion regulation, interactions of emotions with attention and non-cognitive traits like grit, group emotions in collaborative learning, and socio-cultural contextualization to 'social emotions of pride, shame, guilt, jealousy, envy, and so on' (D'Mello 2017, 122).

As noted earlier, smart classrooms may or may not integrate with the smart campus. According to the data imaginary, it makes sense for universities to integrate data across many dimensions of university life because it will allow for richer correlations, classifications, and predictive models. If a student suddenly externalizes emotions such as sadness, boredom, or distress inside of a classroom, it could also help to know if that same student stopped attending social events, eating at the dining hall, and the like. If a smart university surveillance system could study these patterns of behavior, it may be able to flag warning signs of failure or attrition and intervene early to improve performance and throughput rates.

In December 2019, *The Washington Post* reported that dozens of US universities are already moving in this direction. Bluetooth beacons and WiFi access points are being used to surveil the location of students and evaluate a wide variety of behaviors. The data collected is used to monitor attendance and assign student 'risk scores' based on factors like the frequency of visits to the library. They are also integrating Big Data analytics so that students can be assessed according to group characteristics like 'students of color' and 'out-of-state students' (Harwell 2019).

As more data is collected, more accurate and richly detailed evaluations become possible. By tracking visits to the cafeteria or gym, schools can determine 'signs of personal anguish' such as student isolation or eating disorders. Professors can evaluate attendance and tardiness to adjust grades, and 'schools can "take action" against truant students, such as grabbing back scholarship funds' (Harwell 2019).

However, it's worth noting that most universities do not have smart classrooms packed with cameras and IoT devices, especially in the physical classroom space. Chinese deployments have received some attention in the Western press, yet most Chinese classrooms do not have smart technologies (Yang et al. 2018). Google AltSchools developed a system called 'AltVideo' to harness audio and video data that would 'listen to every word a person is saying,' but there are no published results (Lapowsky 2015). Other examples are scattered across countries like India, Australia, Spain, and the United States.

The logic of the smart classroom might sound appealing on the surface: why not incorporate a system that detects boredom or frustration during a lecture, so that a teacher can improve her delivery? However, the alleged benefits of smart classroom technologies remain theoretical. The literature we reviewed did not provide reproducible evidence that smart tech improves educational outcomes at scale across a broad spectrum of

deployments. Yet the use of smart classroom surveillance poses major ethical dilemmas that calls the smart classroom concept into question.

Ethical considerations

Scholars have long recognized ethical issues in education projects driven by computer analytics (Slade and Prinsloo 2013; Watters 2014; Williamson 2015; Sclater 2016; Kwet 2017; Prinsloo and Slade 2017). Upon launching the field of affective computing, Rosalind Picard (1997) observed that computers could ‘potentially build up models’ of our emotional lives ‘over long periods of time, and these models might be particularly sought after, e.g. in lawsuits, insurance matters, and by prospective employees’ (118). If accuracy keeps improving, Picard warned, new capabilities such as lie detection could become problematic if they are imposed on society. She also worried that politicians could use the data to manipulate the public. When equipped with emotion recognition technology, advertisers, marketers, employers, and even potential lovers might likewise alter people’s behavior ‘by subliminal influences and by overt requirements to engage in tasks believed to lead to certain emotions’ (123).

Two decades later, these warnings seem prescient, and they apply to other forms of data analytics, such as the use of IoT devices for location surveillance and video analytics for behavioral analysis. Let us consider several major problems smart classroom technologies pose to students, teachers, and the pedagogical process.

Chilling effects and student performance: Most would agree that free speech and inquiry is critical to education and democracy. Yet studies have shown that people conform to the status quo when they know they are being watched (Penney 2016, 2017). Smart classroom technologies look for active engagement through gestures, activity, and so on, but there is educational value to silent, reflective practices and quiet contemplative spaces. A classroom that detects and scrutinizes every single movement can stifle independence and induce self-censorship among students and teachers, who strive to meet the expectations of the all-seeing machines – and the humans who operate them (Kwet 2017).

Discrimination: Studies have shown that Big Data systems frequently feature algorithmic bias whereby a computer system creates unfair or discriminatory outcomes that have a disparate impact on oppressed and marginalized communities (e.g. Noble 2018; O’Neil 2016; Eubanks 2018). This is partly due to the fact that biased data leads to biased assessments. Smart classrooms may generate algorithmic biases if the data (e.g. physical movements and emotional expressions) of ‘high-achieving’ students and teachers (e.g. upper-class, white, male) differ from those of ‘low-achieving’ students and teachers.

Indeed, a recent study found that emotion-reading technology fails the racial bias test (Rhue 2018). Comparing two different facial recognition emotional analysis services, Face and Microsoft’s Face API, the study found black males were assigned more negative emotions than white males, even when controlling for the degree of smiling. This suggests the use of emotion recognition technologies inside the classroom may have a disparate impact on people of color.

Police and government surveillance: Many school systems are blanketing their campuses with CCTV cameras (Herold 2019), which can be used for both ‘safety and security’ and

smart university applications. CCTV and IoT data can be accessed by law enforcement agencies or used to profile students and teachers deemed threatening. Studies show that CCTV cameras for safety and security have chilling effects on students (Kwet 2019b), and a recent study has shown that policing in schools can negatively impact the educational performance of minority youth (Legewie and Fagan 2019). The constant monitoring of physical activity could therefore create anxiety about police and government surveillance among student and teacher activists, people of color, immigrants, and other marginalized or vulnerable groups.

Technocratic control: Smart classrooms add to the datafication of university affairs. The pervasive quantification of each and every little movement and gesture opens the door to real-time and long-term regimes of technocratic control whereby actors in the educational system are treated as objectified parts in a larger machine of bureaucracy. Education thus becomes an act of reward and punishment according to metrics which classify, predict, and sort learners into various roles for the job market to satisfy the interests of states and corporations (Williamson 2012; Kwet 2017, 2019b).

Waste of resources: Smart classroom advocates argue that their technologies will benefit classrooms, but there is no evidence of any improvements to scale thus far. While projects are in their early stages, it is possible that smart classrooms will do little or nothing to benefit the education system. As Kazilek (2019) remarked, the projects cost money. Classrooms of the future may turn out to waste scarce resources.

Corporate colonization of education: Private ownership and control of classroom technology intrudes upon student and teacher autonomy (Zeide 2017). Smart classrooms powered by Big Tech corporations, privatize core functions like assessment, evaluation, and curriculum development in public education. If proprietary systems are implemented in schools, many classroom experiences will be dictated by the corporations designing them (Kwet 2017, 2019b).

Mission creep: Smart classrooms offer researchers a place to experiment with new forms of surveillance monitoring technologies that can be deployed in other domains, such as crowd control, workplace management, and retail. The ability to monitor emotions accurately with face recognition or correlate data about location with other patterns of life could be used to manipulate workers, crowds, or consumers, and it could threaten the civil rights and liberties of those being monitored.

Authoritarianism and democracy: It is quite possible that smart classroom technologies are applied in a coercive or deceptive fashion within the classroom, without proper disclosure to education participants (Slade, Prinsloo, and Khalil 2019). Students cannot opt out of many technologies, such as CCTV cameras and IoT sensors, while others may be imposed on students and teachers through coercive means, such as a university app that bundles location services for on-campus activities with attendance tracking. Pressure could also be applied by employers expecting an 'optional' dossier of student activity during their time on campus. Students should have the right to completely reject smart university surveillance.

Tech solutionism: Higher education is beset by inequality and fierce competition between universities. A focus on deploying the latest technologies in school distracts attention away from deep-seated socio-economic problems leading to poor and unequal educational outcomes within and between countries.

Conclusion

There are several additional issues which lead us to believe current smart classroom initiatives cannot be ethically justified. First, there is the prospect of turnkey totalitarianism. When infrastructure is put in place that *can* collect personal data, even if monitoring devices are not *currently* doing it, educational participants are threatened by the prospect of ubiquitous surveillance. For example, if Bluetooth beacons are installed in classrooms for a student art exhibit, that same technology could later be used for tracking student behaviors with an official campus app. Once mass surveillance infrastructure is installed, the key is already in the ignition, and totalitarian surveillance is ready to be turned on. The dystopian university is a mouse click away.

Second, there is no practical way to individually opt out of smart classroom surveillance. Even if an *individual* could be excluded from *personal* data collection, aggregated surveillance can still harm teachers and students. Let's say a video camera monitors how often students doze off without collecting individual names. Teachers may use that data to discipline the group, while administrators may use it to discipline the teachers. This outcome could create an unhealthy sensation of moment-to-moment micromanagement ill-suited to education.

Third, even 'privacy-friendly' sensors could pose problems to education. For example, infrared sensors might offer a 'privacy-preserving' alternative to video analytics for applications like body counting to coordinate classroom bookings. However, the first problem (turnkey totalitarianism) is in effect if the data can be used in conjunction with personal data, and the second problem (aggregated micro-management) could also be in effect in some cases.

We recommend that any smart classroom solution which imposes physical surveillance devices on classrooms should be banned. Mass surveillance violates the rights of individuals, chills freedom of expression, undermines dissent, and puts marginalized communities at risk.

Strong privacy laws are needed to protect students and teachers from the encroachment of surveillance and the corporate-driven technological overhaul of classroom spaces. Laws must prevent coercive terms and conditions that allow data controllers to impose surveillance on data subjects.

In its 2019 annual report, research institute AI Now recommended that government regulators ban the use of affect recognition technology for important decisions about human lives – including student performance in school – due to its contested scientific foundations and potential to amplify discrimination (Crawford et al. 2019, 6, 51). We take this recommendation further and propose a hard ban on affect recognition for all educational activity, *even if the technology is accurate*, because it violates educational freedom and feeds into a broader industrial-scale push towards a total surveillance society.

It is also worth noting that technology could be constructive in the classroom without resorting to surveillance and control. Strong Free and Open Source Software policy preferences combined with preferences for federated, decentralized Internet services should be explored by policymakers. Governments can help by subsidizing the development and maintenance of public interest technology, or 'People's Tech,' to push the tech ecosystem towards digital socialism.²

Technologies respecting the freedom of teachers and students should be developed and deployed without imposing tools of surveillance and corporate control. The Free and Open Source Software community provides an array of options for core infrastructure and functions, such as operating systems (GNU/Linux), office suites (LibreOffice), anonymous web browsing (Tor), communications services (Jitsi, Signal), and social networking (Mastodon, Fediverse).

Smart classroom developers should consider using local cloud solutions like Freedom-Box and privacy-minded virtual assistants like Almond. However, it should be noted that freedom-respecting technology cannot always protect classroom participants. In the context of hierarchical power relationships in schools and universities, administrators and educators might still force students and teachers to use devices that spy on them or exploit them, even if those technologies are designed for end-user ownership and control. A shift towards less hierarchical and more democratic models of education would help ensure the technology used serves the interests of classroom participants.

Lastly, school systems should equip their students with the tools to think critically about the digital ecosystem. Students and teachers should collectively determine what a ‘smart’ classroom should be, and what role technology should have in twenty-first century classrooms.

Notes

1. In this paper, we use the term *smart* in a colloquial sense to denote that which is computer-driven. The notion of a ‘smart’ technology lacks a widely accepted or evenly applied definition. A smartphone, for example, is a multi-purpose device typically used for traditional phone functions, Internet services, and multimedia applications. A ‘smart’ surveillance camera, on the other hand, may be considered ‘smart’ because it can evaluate moving images using AI and feed its ‘knowledge’ across a network.
2. See People’s Tech for People’s Power (Kwet [forthcoming](#)); Kwet [2017](#), [2019b](#).

Disclosure statement

No potential conflict of interest was reported by the author(s).

ORCID

Paul Prinsloo  <http://orcid.org/0000-0002-1838-540X>

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